When the Candidate Set Knows the Answer:
Exchangeable Hard Negatives for Vision–Language Evaluation

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The overlooked unit of evaluation is the entire candidate family: a generator can mark its source through relational structure before an image is observed.

0.863
CREPE blind PairAcc

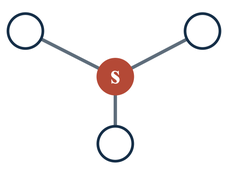
0.000
attack gain after matching

RELATIONAL LEAKAGE → SYMMETRY-CONSTRAINED GENERATION → IMAGE-CONDITIONED EVALUATION

a Candidate incidence

Three binary semantic interventions

	color	pattern	transp.
source	0	0	0
edit 1	1	0	0
edit 2	0	1	0
edit 3	0	0	1



Unit-edit edges shown

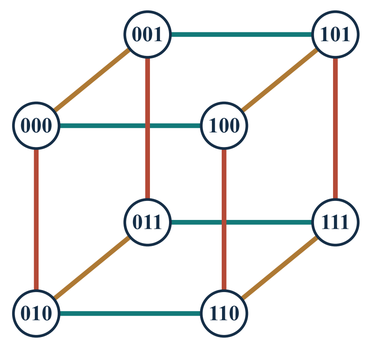
Distance-shell counts

h(source) = (3, 0, 0)

h(edit) = (1, 2, 0)

b Complete the intervention orbit

$C = A_1 \times A_2 \times A_3$, $|A_j| = 2$



0 / 1: black / red; plain / striped; opaque / transparent

color pattern transparency

c Verify every structural role

Distances from each candidate

node	d = 1	d = 2	d = 3
000	3	3	1
001	3	3	1
010	3	3	1
011	3	3	1
100	3	3	1
101	3	3	1
110	3	3	1
111	3	3	1

h(v) = (3, 3, 1) for all v

d Separate two channels

CANDIDATES ONLY

Typed graph / lexical graph / length

Blind PairAcc = 0.500

IMAGE + CANDIDATES

Region ranking or full-image detection

Visual method gains survive

EXPLOIT THE LEAKED ROLE

$a(x,c) = \text{pctRank}(s_{CLIP}) + \text{pctRank}(f_{iso})$

Frozen ranks; equal weights; no encoder update.

→

INTERVENE ON STRUCTURE

Find induced subsets with identical complete distance signatures.

→

FALSIFIABLE PREDICTION

Tied graph ranks → fusion preserves visual ordering → zero attack gain.

Main figure. Three-slot illustration: incidence → full orbit → identical distance shells → separated audit and visual channels. Shell counts illustrate the structural role; the registered empirical attack uses nearest isolation.

01 What leaks, and why?

An item is (image x, candidates C, positive y). Build G(C) from texts alone, without images, model scores, option order or labels.

Exchangeability removes privileged structural roles under the declared interventions.

$\text{Aut}_{\text{typed}}(G)$ acts transitively on V(G).

Every node can be mapped to every other while preserving typed intervention distances.

A constructive symmetry guarantee

XOR with a = u XOR v maps u to v and preserves every typed coordinate difference. Invariant graph-role statistics therefore tie.

Transitivity is sufficient. Length and lexical signatures require separate checks.

Image-free audit on released collections

Collection	Released	Exact
FG-OVD typed	0.701	**0.500**
CREPE lexical	0.863	**0.500**

FG-OVD: 2,466 / 3,005 parsed (82.1%). CREPE: 301,915 sets; 2,244,005 candidates. Scores are invariant to option permutations.

02 A frozen attack and exact test

Freeze the statistic before labels. Sum its average-tie percentile rank and the visual rank. Validate on 152 disjoint vocabularies (76 per track).

Backbone	CLIP	Gain [95% CI]
ViT-B/32	0.765	**+.141 [.115, .168]**
RN50	0.764	**+.130 [.105, .154]**
ViT-B/16	0.747	**+.130 [.107, .154]**

Exact test: search largest subsets first; recompute signatures; select a tied witness before revealing labels. Retain only items whose positive is inside.

Attack gain after matching: 0.000
All three backbones; identical visual ordering.

03 ORBIT-VISUAL construction

Generate from FG-OVD scaffolds and admissible alternatives, without image scores or model queries.

01 Parse and isolate slots

Round-trip the typed parser; retain editable spans and fixed text.

02 Select compatible alternatives

Two values per slot, 2–4 slots: color, pattern, transparency. Freeze material.

03 Optimize the feasible assignment

Maximize dimension, minimize source Hamming distance, then use released order.

04 Emit the full Cartesian product

Preserve spans and punctuation; equalize token lengths; reject same-type and fixed-slot collisions.

05 Certify before evaluation

Check uniqueness, 2^k count, slot balance, parsing and identical typed / lexical signatures.

Frozen construction ledger

Quantity	Value
Built / parsed vocabularies	**1,301 / 2,466**
Coverage	**52.76%**
Candidates	**7,608**
Newly composed nodes*	**77.12%**
Typed / lexical / length PairAcc	**0.500 / 0.500 / 0.500**

Orbit dimension: 2 / 3 / 4 slots



*Mean synthetic-node fraction across vocabularies.

Surface quality is part of the construction

Frozen review: 100 stratified vocabularies, 784 generated negatives; 9 inherited source defects, 0 new defects; all edits stay in declared slots.

Material edits caused agreement and compound instability. The final review tests surface preservation, not physical plausibility.

Why generate instead of filtering?

Post-hoc matching retains 41.2% of parsed FG-OVD and 4.06% of CREPE. Complete orbits build in balance and expose semantic distance.

04 Visual conclusions survive

PairAcc: original candidate family → orbit

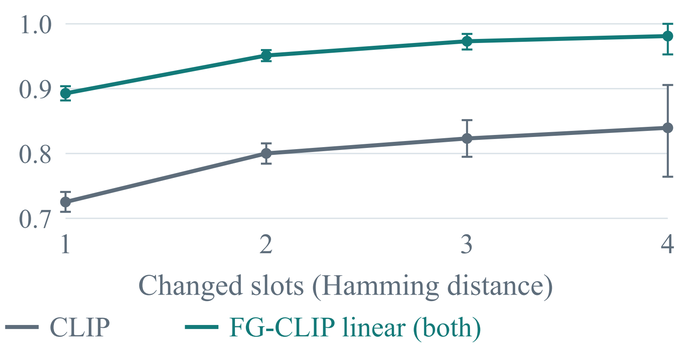
Model	Original	Orbit
CLIP ViT-B/16	0.785	**0.761**
FG-CLIP linear (both)	0.974	**0.917**
FG-CLIP text frozen	0.970	**0.912**
FG-CLIP visual frozen	0.969	**0.913**
FG-CLIP nonlinear	0.974	**0.917**

Region ranking, n = 1,301. All four FG-CLIP projections retain +0.151 to +0.157 gains without training.

Detector	Original	Orbit
OWLv2-B/16	0.842	**0.770**
NoctOWLv2-B/16	0.967	**0.916**

Common tokenizer-valid n = 597. Orbit gain: +0.147 [.121, .172]. Localization recall@.5: 0.980 / 0.973.

One-slot changes are hardest



Region curves; 95% bootstrap CIs. Image shuffle (n = 240): CLIP drops 0.216; FG-CLIP variants drop at least 0.114 (p = .001).

05 Control the number of options

Match orbit counts with three frozen random subsets (n = 595). Compare detector method gains: orbit minus mean same-size control.

Metric	Difference	95% CI
PairAcc	+0.028	**[+.003, +.051]**
Top-1	+0.009	**[-.040, +.058]**
AP50	+0.011	**[-.031, +.052]**
mAP50:95	+0.008	**[-.025, +.041]**

Only PairAcc separation exceeds the count-matched control reliably.

DESIGN PRINCIPLE **Audit the candidate family before modeling; balance interventions at generation time; report semantic distance and matched-count controls.**

Metric: PairAcc averages positive–negative wins within each item, then across items; a tie receives 1/2. Its chance value stays at 0.500 for every option count. Scope: graph-role guarantees apply to the declared intervention family; detector AP is a selected-positive subset metric.

Selected references: FG-OVD (Bianchi et al., 2024); CREPE (Ma et al., 2023); CLIP (Radford et al., 2021); FG-CLIP (Xie et al., 2025); OWLv2 (Minderer et al., 2023); NoctOWLv2 (Bianchi, 2025).
Evidence: frozen ORBIT-VISUAL v5 and paired released-checkpoint evaluations.